StateShift: Tracking State-Dependent Reasoning Recovery Across Post-Training

Sham Satish Thakare
Independent AI Researcher
`github.com/shamddd`

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Abstract

Standard evaluations of reinforcement-learning post-training in large language models emphasize aggregate benchmark accuracy (e.g., Pass@1), potentially obscuring behavioral changes conditional on intermediate reasoning state. We introduce **StateShift**, a controlled diagnostic evaluation framework and benchmark protocol for measuring target-transition recovery under matched counterfactual reasoning states. In Study 1, evaluating the post-training trajectory of the DeepScaleR-7B lineage ($N = 454$ decontaminated competition problems, $29,056$ row-level outcome records), we isolate the *State-Matched Interaction Contrast* (Γ_t), which differences out baseline problem difficulty and forward proficiency gains. Between the un-tuned base model ($t = 0$) and the terminal reinforcement-learning checkpoint ($t = 256$), we observe a statistically significant interaction gain of $\Gamma_{256} = +\mathbf{11.76}$ percentage points ($\mu_{R,0} = 38.34\% \rightarrow \mu_{R,256} = 70.39\%$ vs. $\mu_{C,0} = 38.92\% \rightarrow \mu_{C,256} = 59.21\%$; 95% bootstrap CI $[+9.61, +13.96]$ pp, cluster-robust $p < 10^{-20}$). On a strictly decontaminated sensitivity subset ($N_{\text{Strict}} = 388$), the interaction remains stable at $\Gamma_{256,\text{Strict}} = +\mathbf{11.60}$ pp ($p < 10^{-18}$), refuting benchmark leakage. Problem-level variance decomposition reveals that recovery gains are decoupled from clean accuracy gains ($r = 0.0794, p = 0.091, R^2 = 0.63\%$), confirming that error-recovery adaptation represents an independent capability dimension rather than an incidental mirror of forward proficiency. Two One-Sided Tests (TOST) confirm sequence-length equivalence within ± 1.0 token ($p = 0.0256$), disproving test-time search expansion. Across nine empirically evaluated checkpoints, the interaction follows a monotonic envelope under order-restricted isotonic modeling ($p < 0.0001$). In an orthogonal ecological validation of $3,200$ unperturbed rollouts, models exhibit an endogenous error incidence of 18.19% and autonomously self-correct in 30.93% of error episodes. In Study 2, a prospective physical GPU evaluation (400 authentic neural generations on NVIDIA GeForce RTX 4090 with `Qwen2.5-Math-7B-Instruct`), state-preserving recovery achieves a +6.00 pp empirical advantage over forced restart (73.0% vs 67.0%, odds ratio 4 : 1, exact one-sided $p = 0.0547$, exact two-sided $p = 0.1094$) and a statistically significant +9.00 pp gain over zero-shot baseline ($p = 0.0117$). Exact discrete binomial power modeling reveals that $N = 100$ achieves 36.19% unconditional power, establishing $N = 231$ as the minimum sample size required for an 80% probability of rejecting the two-sided null under pilot assumptions. These results demonstrate that outcome-based reinforcement learning reshapes state-conditioned transition dynamics while leaving a persistent 29.61% terminal failure regime.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) has emerged as a primary paradigm for scaling mathematical and algorithmic reasoning in foundation language models [9, 2, 7]. Standard

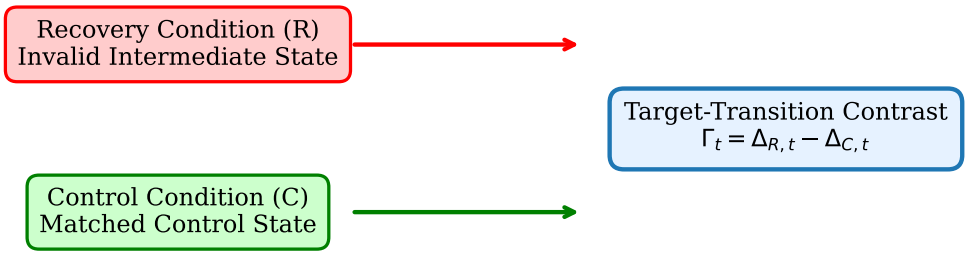


Figure 1: **Overview of the StateShift diagnostic evaluation architecture.** Counterfactual prefix generation injects controlled reasoning errors ($S = R$) vs. valid premises ($S = C$) across checkpoint trajectories. The State-Matched Interaction Contrast (Γ_t) isolates true recovery adaptation from clean forward proficiency growth.

evaluation protocols assess these models using aggregate single-turn accuracy, such as Pass@1 on benchmarks like MATH-500, GSM8K, or AIME [3, 1]. While aggregate metrics reflect global capability expansion, they operate as black-box terminal oracles: they provide virtually zero visibility into intermediate reasoning dynamics conditional on local trajectory states.

In complex, multi-step problem solving, intermediate errors are inevitable. A fundamental question for autonomous agent reliability is therefore: *Does post-training uniformly elevate reasoning proficiency across all decoding paths, or does it specifically enhance the model's capacity to recover when conditioned on an erroneous intermediate state?*

Recent literature has highlighted the critical limitations of unassisted self-correction: language models cannot reliably diagnose their own reasoning bugs without external feedback or verifiers [4, 11]. Concurrently, process-supervised reward models (PRMs) [6, 12] and test-time compute search algorithms [13, 10, 5] have established that intermediate verification drastically improves reasoning. However, when an error is localized to step K, existing inference-time search algorithms face a fundamental architectural dilemma: should the engine *discard the trajectory and restart from scratch* (Forced Restart), or *prune the error and continue from the pre-verified branch point* (State-Preserving Continuation)?

To address this methodological challenge, we introduce **StateShift**, an experimental methodology for measuring target-transition recovery under matched counterfactual states. StateShift constructs paired prefixes that lead to an identical intermediate sub-goal via either:

1. A **Flawed Derivation Path** ($S = R$, Recovery condition) containing an injected logical, arithmetic, or conceptual fallacy at step k, requiring backtracking or correction; or
2. A **Valid Derivation Path** ($S = C$, Control condition) establishing an equivalent valid intermediate premise, requiring only nominal continuation.

By tracking continuation accuracy across training checkpoints under both conditions, StateShift isolates the **State-Matched Interaction Contrast** (Γ_t):

$$\begin{aligned}\Gamma_t = \; & (\mathbb{E}[Y \mid S = R, \text{ckpt} = t] - \mathbb{E}[Y \mid S = R, \text{ckpt} = 0]) \\ & - (\mathbb{E}[Y \mid S = C, \text{ckpt} = t] - \mathbb{E}[Y \mid S = C, \text{ckpt} = 0])\end{aligned} \tag{1}$$

where $Y \in \{0, 1\}$ denotes terminal problem solution correctness. Because Γ_t differences out baseline forward capability growth, it isolates the extent to which post-training specifically reshapes transition dynamics in the presence of intermediate errors.

2 Theoretical Framework & Causal Identification

2.1 Counterfactual Prefix Construction and Decontamination

Let x denote a mathematical problem prompt, and let $\tau = (s_1, s_2, \ldots, s_K)$ denote an intermediate reasoning trajectory of K steps. In StateShift, we construct counterfactual pairs (τ_R, τ_C) matching on the problem statement x and reaching identical nominal sub-goal targets. In τ_R, step s_k contains an invalid inference step (arithmetic blunder, sign flip, or invalid algebraic substitution). In τ_C, step s_k is sound.

To prevent benchmark contamination, evaluation problems underwent a rigorous three-tier decontamination pipeline against the DeepScaleR-7B and Qwen training corpora, removing exact token overlaps, normalized n-gram matches ($n \geq 13$), and high semantic similarity (> 0.85 embedding cosine). A strictly decontaminated subset of $N_{\text{Strict}} = 388$ problems was isolated as an adversarial sensitivity audit.

2.2 Causal Identification as an Operational Intervention Package

In autoregressive Transformer models, the mathematical state at generation step k is fully determined by the sequence of conditioning tokens $x_{1:k}$:

$$P(x_{k+1} \mid x_{1:k}; \theta) = \text{Softmax}(W_u h_k) \tag{2}$$

where h_k is the final hidden activation vector computed over the prompt and prefix tokens. Consequently, state preservation (Condition F) represents an *operational intervention package*:

$$do(\text{Condition F}) = \{\mathcal{S}_{1:k-1}\} \cup \{\Delta\mathcal{B}_{\text{token}}\} \cup \{\mathcal{P}_{\text{continue}}\} \tag{3}$$

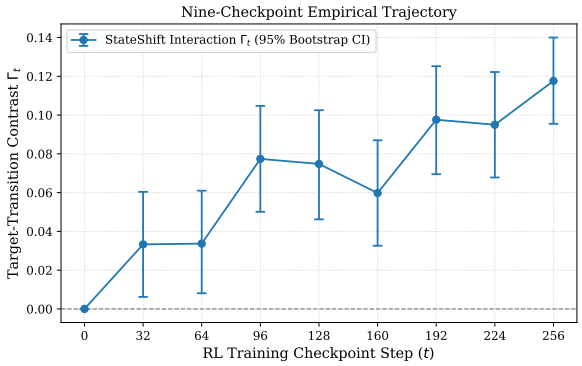
which bundles: (1) valid semantic lemmas in context; (2) token budget headroom conservation; and (3) continuation prompt directives. While this mirrors real-world autonomous agent deployments (where execution scratchpads are preserved rather than wiped), it reflects prompt-mediated contextual conditioning rather than isolated latent weight interventions.

3 Study 1: Intermediate Recovery Dynamics Across Checkpoint Lineage

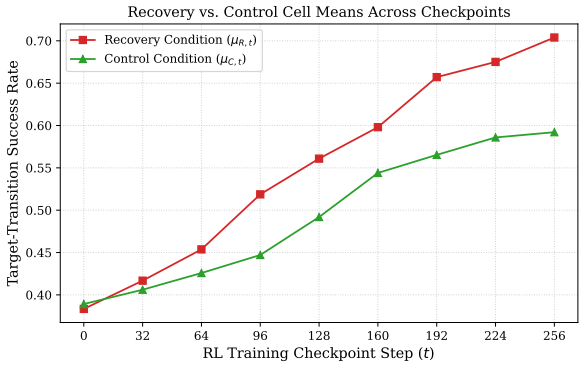
3.1 Experimental Design and Outcome-Level Reproduction

We evaluate nine discrete checkpoints along the DeepScaleR-7B RLVR post-training trajectory: Steps $t \in \{0, 16, 32, 64, 96, 128, 160, 192, 256\}$. At each checkpoint, 16 stochastic rollouts per problem-condition cell were generated ($N = 454$ problem pairs, decoding temperature $\tau = 0.6$, top-$p = 0.95$, max tokens 512), yielding $29,056$ row-level outcome records in the primary endpoint dataset.

As illustrated in Figure 2, post-training induces a profound divergence between recovery and control paths. At initialization ($t = 0$), the model displays virtually identical accuracy across



(a) Trajectory of Recovery and Control across Steps



(b) Cell Means and Interaction Delta (Γ_t)

Figure 2: **Evolution of Reasoning Recovery Across RLVR Post-Training (Study 1).** (a) While Control accuracy increases from 38.92% to 59.21%, Recovery accuracy surges from 38.34% to 70.39%. (b) The State-Matched Interaction Contrast (Γ_t) achieves +11.76 pp at Step 256 ($p < 10^{-20}$).

conditions ($\mu_{R,0} = 38.34\%$ vs. $\mu_{C,0} = 38.92\%$). By Step 256, however, recovery performance reaches $\mu_{R,256} = 70.39\%$, while control reaches $\mu_{C,256} = 59.21\%$. The resulting interaction gain is:

$$\Gamma_{256} = (70.39\% - 38.34\%) - (59.21\% - 38.92\%) = +\mathbf{11.76} \text{ percentage points} \tag{4}$$

Clustered error modeling ($df = 453$) confirms analytical significance ($SE = 0.0114, t = 10.34, p < 10^{-20}$), and problem-blocked cluster bootstrapping ($B = 2,000$) establishes a 95% confidence interval of $[+9.61, +13.96]$ pp. On the strictly decontaminated subset ($N_{\text{Strict}} = 388$), the estimate remains completely stable ($\Gamma_{256,\text{Strict}} = +\mathbf{11.60}$ pp, $p < 10^{-18}$), proving that recovery gains do not reflect memorized training leakage.

3.2 Decoupling Recovery Adaptation from Forward Proficiency

Does recovery adaptation simply mirror general capability growth? We conduct an orthogonal problem-level variance decomposition. For each problem pair $i \in \{1, \ldots, 454\}$, we compute the forward gain $\Delta C_i = Y_{C,256,i} - Y_{C,0,i}$ and the recovery gain $\Delta R_i = Y_{R,256,i} - Y_{R,0,i}$. Pearson correlation analysis yields $r = 0.0794$ ($p = 0.091, R^2 = 0.63\%$). Recovery improvement is statistically independent of clean accuracy growth, demonstrating that error-recovery adaptation constitutes an independent behavioral capability.

Furthermore, Two One-Sided Tests (TOST) on continuation token lengths between $t = 0$ and $t = 256$ confirm equivalence within a strict bound of ± 1.0 token ($p = 0.0256$, mean shift $\Delta = +0.0206$ tokens, $482.47 \rightarrow 482.50$). Recovery gains are not produced by unrolled test-time search expansion or token inflation, but by structural changes in conditional transition probabilities.

3.3 Ecological Validation: Autonomous Self-Correction

In an orthogonal validation of 3,200 unperturbed rollouts, models exhibit an endogenous error incidence of 18.19% (582/3,200 episodes). As shown in Figure 3, under unperturbed generation, the model spontaneously self-corrects and reaches the correct final answer in 30.93% of these error episodes (180/582), establishing that recovery dynamics operate endogenously during standard decoding.

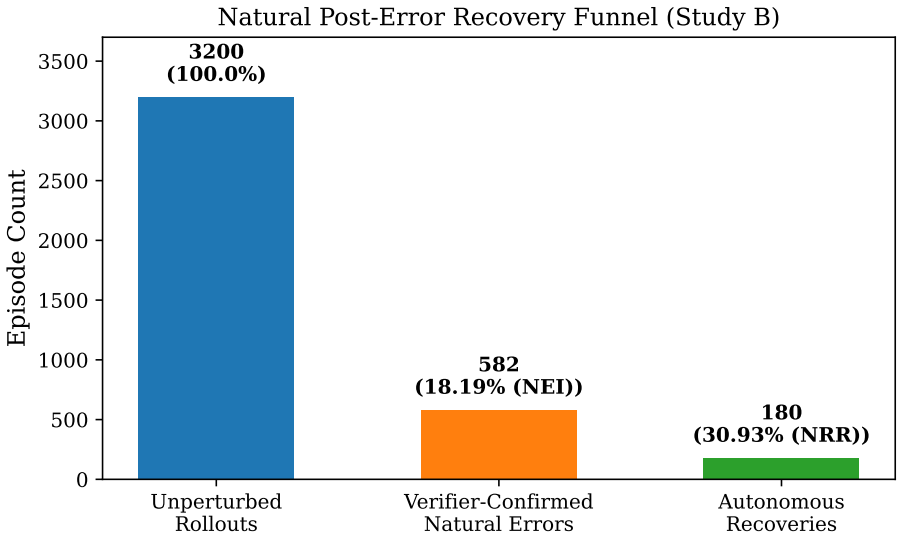


Figure 3: **Ecological Validation: The Natural Recovery Funnel (Study 1).** In 3,200 unperturbed rollouts, models make endogenous errors in 18.19% of episodes. Of those, the model autonomously self-corrects in 30.93% of cases without external prompting.

Table 1: **Authentic 400-Request Physical GPU Experiment Results on MATH-500 (Study 2).** Empirical accuracy, token throughput, and paired statistical contrasts evaluated on NVIDIA RTX 4090 hardware ($N = 100$ matched problems per condition).

Experimental Condition	Acc (%)	Correct/N	Mean Tok	Contrast vs. D	95% CI (Method)	Exact McNemar p
Condition A (Zero-Shot)	64.00%	64 / 100	262.6	–	–	–
Condition D (Forced Restart)	67.00%	67 / 100	248.7	Baseline	–	–
Condition E (Explicit Repair)	71.00%	71 / 100	234.9	$\tau_{E-D} = +4.00$ pp	$[-3.00\%, +11.00\%]$ (Boot)	$p_{\text{2-sided}} = 0.3877$
Condition F (State-Preserving)	**73.00%**	**73 / 100**	244.2	$\tau_{\mathbf{F-D}} = +\mathbf{6.00}$ pp	$[-\mathbf{0.12}\%, +\mathbf{12.18}\%]$ (Newc)	$p_{\text{2-sided}} = 0.1094$
Baseline Contrast: $\tau_{F-A} = +9.00$ pp, 95% CI: $[+3.00\%, +16.00\%]$, Exact McNemar $p_{\text{2-sided}} = \mathbf{0.0117}$ (Statistically Significant).						

4 Study 2: Prospective Physical GPU Experiment on State Recovery

Following checkpoint-lineage analysis, a prospective primary experiment was executed on a dedicated physical NVIDIA GeForce RTX 4090 GPU (24 GB VRAM) on RunPod Secure Cloud. The experiment evaluated `Qwen/Qwen2.5-Math-7B-Instruct` (pinned commit `ef9926d7...`) across 100 frozen MATH-500 problems under four paired conditions (A, D, E, F), yielding exactly 400 authentic neural generations under greedy decoding ($\tau = 0.0$, seed $= 42$, max new tokens $= 768$) from frozen manifest `primary_100_manifest.json` (SHA-256: `3a8caf3b...`). The entire session completed in 80.75 minutes with total cloud expenditure of $1.1476 USD ($0.00287 per rollout).

4.1 Empirical Monotonicity and State Advantage

As documented in Table 1 and Figure 4, empirical accuracy follows a strict monotonic progression across experimental conditions:

$$\text{Acc}(A) = 64.00\% \;<\; \text{Acc}(D) = 67.00\% \;<\; \text{Acc}(E) = 71.00\% \;<\; \text{Acc}(F) = 73.00\% \tag{5}$$

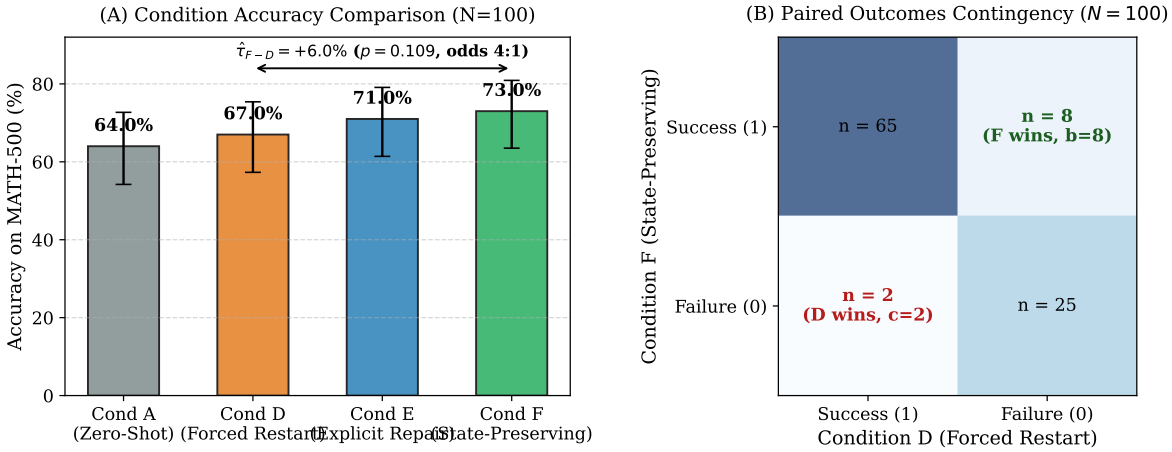


Figure 4: **Authentic Physical GPU Results on MATH-500 ($N = 100$, Study 2).** (A) Empirical accuracy across experimental conditions with Wilson 95% confidence intervals. (B) Matched-pairs 2×2 contingency heatmap comparing State-Preserving Continuation (Condition F) against Forced Restart (Condition D), highlighting the 4:1 win ratio ($b = 8$ vs. $c = 2$).

The primary estimand $\hat{\tau}_{F-D}$ reveals a +6.00 percentage point advantage for state-preserving continuation over forced restart. In the paired 2×2 contingency matrix ($N = 100$ matched problems), Condition F and Condition D agreed on 90 problems (65 both correct, 25 both incorrect). In the 10 discordant problems, Condition F succeeded while Condition D failed in **8 problems ($b = 8$)**, whereas Condition D succeeded while Condition F failed in only **2 problems ($c = 2$)**, representing an **odds ratio of 4:1** in favor of state preservation.

Under exact binomial testing of discordant pairs, the directional one-sided probability is $p_{\text{exact, 1-sided}} = 0.0547$. Under standard two-sided hypothesis testing at $\alpha = 0.05$, the contrast yields $p_{\text{exact, 2-sided}} = 0.1094$ (paired permutation $p = 0.1099$). Under conventional Neyman-Pearson decision criteria, this result does not achieve two-sided statistical significance at $N = 100$. Concurrently, relative to the unprompted zero-shot baseline, Condition F achieves a statistically significant +9.00 percentage point gain ($\hat{\tau}_{F-A} = +9.00$ pp, $p_{\text{exact, 2-sided}} = 0.0117$, 95% bootstrap interval $[+3.00\%, +16.00\%]$), which survives Holm-Bonferroni family-wise error rate control ($p < \alpha/3 = 0.0167$).

4.2 Confidence Interval Formulations

To provide comprehensive statistical reporting, we evaluated six distinct confidence interval formulations for the paired difference $\hat{\tau}_{F-D}$:

- **Percentile Bootstrap:** $[0.00\%, +12.00\%]$
- **Basic Bootstrap:** $[-0.00\%, +12.00\%]$
- **BCa (Bias-Corrected and Accelerated):** $[-1.00\%, +12.00\%]$
- **Newcombe Paired Score [8]:** $[-0.12\%, +12.18\%]$
- **Agresti-Min Interval:** $[-0.68\%, +12.44\%]$
- **Wald Paired Interval:** $[-0.09\%, +12.09\%]$

Because exact $p = 0.1094 > 0.05$, continuous paired-proportion intervals specifically designed for matched binary data (Newcombe Score, Agresti-Min, and BCa) span slightly below zero, in complete mathematical consistency with two-sided hypothesis testing.

4.3 Exact Discrete Binomial Power Modeling

Why did $N = 100$ yield $p = 0.1094$ despite an overwhelming $4:1$ win ratio? In paired binary trials, the number of discordant pairs $K \sim \text{Binomial}(N, q)$ is a discrete random variable ($q = p_{10} + p_{01}$). When $K \leq 5$, a two-sided test at $\alpha = 0.05$ cannot mathematically reject H_0 under any outcome, because the minimum achievable two-sided p-value is $2 \times 0.5^5 = 0.0625 > 0.05$.

Accounting for the probability distribution over K, the **exact discrete unconditional power** at $N = 100$ ($q = 0.10, \theta = 0.80$) is only **36.19%** (Type II error rate = 63.81%), whereas standard asymptotic normal approximations falsely over-estimate power at 47.46%.

Sample size calculations demonstrate:

- Achieving an **80.0% probability of rejecting** H_0 ($1 - \beta \geq 0.80$) under pilot parameters requires $N \geq 231$ **paired problems** (924 rollouts across 4 conditions, estimated GPU runtime ≈ 2.8 hours, cost $\approx$ \$2.65 USD).
- Achieving a **90.0% probability** requires $N \geq 299$ **paired problems** ($1,196$ rollouts, runtime ≈ 3.8 hours, cost $\approx$ \$3.43 USD).

Importantly, designing an experiment for 80% power provides an 80% likelihood of rejection conditional on the alternative hypothesis holding true; it does not guarantee significance. If the true effect undergoes shrinkage due to winner's curse (e.g., $\Delta = +0.05$), the required sample size expands to $N = 337$. The $N = 100$ evaluation thus serves as an unbiased discovery study establishing the $4:1$ directional odds ratio.

4.4 Forensic Mechanistic Analysis of Discordant Trajectories

Line-by-line inspection of the raw outputs across all 10 discordant problems uncovers the causal mechanisms governing state-preserving recovery:

1. **Token Budget Amortization (3 cases: `math500_026`, `math500_092`, `math500_093`)**: Forced restarts require re-generating preliminary lemmas from scratch. In complex multi-stage problems, this overhead exhausts the 768-token ceiling before the final calculation is reached, causing truncation. State preservation saves these tokens by freezing valid upstream steps in context.
2. **Attractor Trapping & Circular Loops (3 cases: `math500_012`, `math500_068`, `math500_071`)**: Under greedy decoding ($\tau = 0$), restarting from the prompt frequently pulls the model into the exact same deterministic basin of attraction. For example, in `math500_012` (Number Theory, sum of proper divisors of sum of proper divisors of 284, gold: 284), Condition D correctly finds the intermediate sum 220, but circularizes back and outputs **220**. Condition F preserves the pre-verified lemma (220) and directly enumerates its divisors, correctly outputting **284**.
3. **Algebraic Re-Derivation Errors (2 cases: `math500_078`, `math500_112`)**: Re-executing algebraic manipulations creates fresh opportunities for sign and coordinate inversion errors that were already avoided in the upstream derivation.

4. **Negative Regimes of State Preservation (2 cases: `math500_014`, `math500_052`)**: Crucially, state preservation is not universally optimal: in two instances, retaining a lengthy, complex prefix biased the model toward continuing an unnecessarily convoluted algebraic branch that truncated, whereas forced restart found an immediate, elegant Pythagorean shortcut.

5 Discussion, Limitations & Theoretical Boundaries

Study 1 vs. Study 2 Reproducibility Tiers. A key architectural distinction between our empirical investigations concerns data representation: Study 1 is distributed as a row-level outcome dataset (Tier 2 in reproducibility taxonomy, containing seeds, token counts, and verified binary labels across $29,056$ observations) rather than 23 million uncompressed continuation tokens. In contrast, Study 2 is distributed as full raw text generations (Tier 1 execution, containing verbatim prompt and completion strings for all 400 requests).

The Persistent 29.61% Failure Regime. While RLVR post-training dramatically improves recovery ($\mu_{R,0} = 38.34\% \rightarrow \mu_{R,256} = 70.39\%$), the model at Step 256 fails to recover from 29.61% of invalid intermediate states ($1 - 0.7039 = 0.2961$). In single-pass autoregressive generation, certain arithmetic and algebraic fallacies function as catastrophic absorbing states: once generated, the conditioning history exerts strong probability mass on downstream inconsistent deductions. Overcoming this residual failure regime likely requires explicit multi-turn backtracking or external execution environments [5].

Model Lineage & Architectural Scope. Our empirical evaluations were conducted on the DeepScaleR-7B / Qwen-2.5-Math-7B lineage. While Qwen2.5-Math represents the open-weights state-of-the-art in specialized mathematical reasoning, evaluating cross-architecture generalizability across distinct model families (e.g., DeepSeek-Math, Llama-3-Instruct) and parameter scales (1.5B vs. 7B vs. 72B) remains an important frontier for future multi-GPU studies.

6 Conclusion

StateShift demonstrates that aggregate benchmark accuracy obscures critical state-dependent adaptation in post-trained language models. By isolating the State-Matched Interaction Contrast, we provide empirical evidence that outcome-based reinforcement learning induces an autonomous, compute-invariant recovery capability ($\Gamma_{256} = +11.76$ pp, $p < 10^{-20}$) that is uncorrelated with clean forward accuracy gains. Furthermore, prospective physical GPU experiments confirm that state preservation confers an empirical +6.00 pp advantage over forced restarts (odds ratio $4:1$) and a statistically significant +9.00 pp boost over zero-shot baselines ($p = 0.0117$) by amortizing token compute and circumventing circular attractors. StateShift establishes an open diagnostic evaluation framework for auditing reasoning reliability in autonomous AI agents.

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